Against a Universal Trading Strategy: No-Arbitrage, No-Free-Lunch, and Adversarial Counterpaths

Karl Svozil [1]

[1] *Institute for Theoretical Physics, TU Wien, Wiedner Hauptstrasse 8-10/136, A-1040 Vienna, Austria* [*]

(Dated: July 27, 2026)

We investigate universally winning trading strategies, defined as admissible, self-financing strategies whose discounted gain from zero initial capital is strictly positive on every admissible market trajectory. Such a strategy is a classical arbitrage, and its pathwise requirement is stronger than the usual almost-sure condition. It is therefore excluded already by no-arbitrage and, a fortiori, in locally bounded semimartingale markets satisfying No Free Lunch with Vanishing Risk and admitting an equivalent local martingale measure. Separate modeling regimes yield complementary, but logically distinct, limitations. In a uniform binary prediction model, every causal predictor has mean directional accuracy 1/2; optional stopping nevertheless permits an arbitrarily high trade win rate while preserving zero expected gain. In a worst-case computational model, every total deterministic trading algorithm has a computable adaptive counterpath on which its gain is non-positive. Ville's inequality and computable randomness connect the probabilistic and computational regimes without requiring an adaptive adversary. We make the thermodynamic analogy precise only at the level of positive density martingales and fluctuation identities, rather than identifying a martingale measure with detailed balance or invoking financially relevant Landauer costs. Finally, an explicit put–call-parity analysis of the Wheel Options Strategy shows that its apparent income is compensation for put-like downside and capped upside. Positive results such as Cover's universal portfolio are shown to concern relative regret, not guaranteed absolute profit.

I. INTRODUCTION

The question of whether a trading strategy can always make money, regardless of what the market does, is older than modern mathematical finance. The mathematical study of unpredictable prices begins with Bachelier's 1900 thesis [1], continues through Samuelson's martingale formulation [2], and informs Fama's Efficient Market Hypothesis (EMH) [3]. The EMH is an economic claim about the incorporation of information into prices; it is not, by itself, a theorem that no individual strategy can win unconditionally.

Here a strategy is called *universal* only if, after discounting by a specified numeraire, it produces a strictly positive terminal gain from zero initial capital on every admissible market trajectory. The zero-cost, self-financing, and discounted-gain qualifications are essential: a positive initial investment in a bank account has positive terminal wealth, but it is not an arbitrage and does not outperform its financing benchmark.

This paper articulates the resulting impossibility through three distinct theoretical paradigms, with their assumptions kept separate:

1. **Financial (Measure-Theoretic, Sec. III).** Pathwise universality is a special case of classical arbitrage. It is excluded by no-arbitrage and, under standard semimartingale hypotheses, by the existence of an equivalent local martingale measure. Optional stopping also explains why a high fraction of profitable trades need not imply positive expected gain.
2. **Combinatorial (Uniform Prediction, Sec. V).** The Wolpert–Macready No-Free-Lunch (NFL) theorem concerns uniform averaging over finite objective-function spaces [4]. For trading-direction claims we prove the relevant statement directly: averaged uniformly over all binary price-direction sequences, every causal predictor has accuracy 1/2.
3. **Computational (Adaptive Counterpaths, Sec. VI).** Against every total deterministic computable strategy, a computable environment can simulate its announced position and choose the next return with the opposite sign. A separate martingale-randomness result describes what happens under a passive computable fair measure.

We frame these findings with a *time-reversal stress test* (Sec. II) and a carefully limited Maxwell's Demon analogy (Sec. IV). The social-choice comparison in Sec. VII is explicitly analogical rather than a reduction of one theorem to another. The Wheel Options Strategy is analyzed in Sec. VIII, Cover's different notion of universality is separated from absolute profit in Sec. IX, and the conditional scope of strategies working *for all practical purposes* (FAPP) is discussed in Sec. X.

II. THE TIME-REVERSAL STRESS TEST

Let $\mathcal{M}$ be a space of strictly positive paths $P\colon [0,T] \to \mathbb{R}^n_{>0}$. The time reversal of P is

$$\widetilde{P}(t) = P(T-t), \qquad t \in [0,T]. \tag{1}$$

[*] karl.svozil@tuwien.ac.at; http://tph.tuwien.ac.at/~svozil

For a causal strategy σ, let $\Pi_T[\sigma; P]$ denote its terminal discounted gain from zero initial capital when evaluated on P.

Definition 1 (Pathwise Universal Strategy)**.** A causal strategy σ is *pathwise universal* on $\mathcal{M}$ if it is self-financing and admissible and satisfies $\Pi_T[\sigma; P] > 0$ for every $P \in \mathcal{M}$.

Criterion 1 (Reversal Falsification Test)**.** Suppose $\mathcal{M}$ is closed under time reversal. If there exists $P \in \mathcal{M}$ such that

$$\min\{\Pi_T[\sigma; P], \Pi_T[\sigma; \widetilde{P}]\} \leq 0, \tag{2}$$

then σ is not pathwise universal on $\mathcal{M}$.

Proof. Both P and $\widetilde{P}$ belong to $\mathcal{M}$. A non-positive gain on either member of the pair contradicts Definition 1. □

Criterion 1 is deliberately a falsification device, not an independent impossibility theorem. Reversing a profitable path does not in general negate the profit of a causal strategy, because the strategy can take different positions on the reversed history. The value of the test is that it generates a paired stress scenario whenever the modeled path class is reversal-closed.

Remark 1 (Filtrations and Time Reversal). In stochastic finance, the set of admissible trajectories is not automatically closed under time reversal. The process $P(T-t)$ is generally not adapted to the original forward filtration, and its semimartingale property under a reversed filtration requires additional hypotheses. The stress test is therefore pathwise; the rigorous probabilistic constraint is given next.

III. THE NO-ARBITRAGE FOUNDATION

A. Discounted Gains and Admissibility

Let the market be defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{P})$. Let $B_t > 0$ be a numeraire and S_t the vector of traded asset prices. Write $\overline{S}_t = S_t/B_t$ for discounted prices. A predictable, $\overline{S}$-integrable strategy H has discounted self-financing gain

$$G_t = (H \cdot \overline{S})_t, \qquad G_0 = 0. \tag{3}$$

It is *admissible* if there is a finite constant a such that

$$G_t \geq -a \quad \text{for all } t \in [0,T], \quad \mathbb{P}\text{-a.s.} \tag{4}$$

This uniform lower bound rules out doubling systems financed by unbounded credit.

Theorem 1 (Universality Implies Classical Arbitrage)**.** *If an admissible self-financing strategy H with zero initial capital satisfies $G_T(\omega) > 0$ for every $\omega \in \Omega$, then H is a classical arbitrage.*

Proof. The pathwise assumption implies $G_T \geq 0$ almost surely and $\mathbb{P}(G_T > 0) = 1$. Together with zero initial cost, self-financing, and admissibility, these are stronger than the usual classical-arbitrage conditions $\mathbb{P}(G_T \geq 0) = 1$ and $\mathbb{P}(G_T > 0) > 0$. □

Remark 2 (Terminology and Strength). The pathwise condition is sometimes called a sure or strong arbitrage, but it is not synonymous with *arbitrage of the first kind*. The latter is the condition dual to No Unbounded Profit with Bounded Risk (NUPBR/NA1) and is a distinct, weaker viability concept [5]. Classical no-arbitrage alone already excludes the universal strategy in Theorem 1; NFLVR is invoked below because it supplies the standard martingale-measure formulation.

B. The Martingale Constraint

For a locally bounded semimartingale price process, the Delbaen–Schachermayer form of the First Fundamental Theorem states that No Free Lunch with Vanishing Risk (NFLVR) is equivalent to the existence of an equivalent local martingale measure (ELMM) $\mathbb{Q} \approx \mathbb{P}$ under which $\overline{S}$ is a local martingale [6, 7].

Corollary 2 (No Universal Zero-Cost Strategy)**.** *If an ELMM $\mathbb{Q}$ exists, no admissible self-financing strategy with zero initial capital can have $G_T > 0$ on every path.*

Proof. Under $\mathbb{Q}$, the admissible stochastic integral $G = H \cdot \overline{S}$ is a local martingale bounded from below and therefore a supermartingale. Hence $\mathbb{E}_{\mathbb{Q}}[G_T] \leq G_0 = 0$. If $G_T > 0$ on every path, then equivalence gives $G_T > 0$, $\mathbb{Q}$-a.s., and consequently $\mathbb{E}_{\mathbb{Q}}[G_T] > 0$, a contradiction. □

The ELMM conclusion concerns discounted gains under the pricing measure $\mathbb{Q}$. It does not say that every risky strategy has non-positive expected return under the physical measure $\mathbb{P}$; positive physical expected returns may compensate risk.

C. Win Rate Is Not Expected Value

The probability of closing a trade at a profit can greatly exceed $1/2$ even in a fair market. Optional stopping gives the precise distinction.

Proposition 3 (Asymmetric Barriers in a Fair Random Walk)**.** *Let X_n be a simple symmetric random walk with $X_0 = 0$, and for positive integers a, b let*

$$\tau = \inf\{n \geq 0 : X_n = a \text{ or } X_n = -b\}.$$

Then

$$\mathbb{P}(X_\tau = a) = \frac{b}{a+b}, \qquad \mathbb{E}[X_\tau] = 0. \tag{5}$$

Proof. The stopped process $X_{n\wedge\tau}$ is a bounded martingale, so Doob's optional-stopping theorem applies [8]. If $p = \mathbb{P}(X_\tau = a)$, then $0 = \mathbb{E}[X_\tau] = ap - b(1-p)$, which gives $p = b/(a+b)$. □

With a take-profit of $a = 1$ and a stop-loss of $b = 9$, the win probability is 0.9, but the expected payoff is zero: nine frequent unit gains are offset by one nine-unit loss in expectation. Thus prediction accuracy, trade win rate, and expected discounted profit are three different quantities.

IV. THE MAXWELL'S DEMON ANALOGY: A DENSITY-MARTINGALE BOUNDARY

A. Formal Analogy, Not Equivalence

Maxwell's Demon observes microscopic degrees of freedom and uses information to extract work from fluctuations. A trading algorithm observes a signal and uses it to choose a position. The comparison is useful only after the equilibrium constraints on both systems are stated explicitly (Table I).

TABLE I. A limited analogy between information engines and trading strategies.

Information engine	**Trading strategy**
Observes microscopic data	Observes prices or signals
Uses feedback to extract work	Uses positions to seek gain
Requires cyclic accounting	Requires self-financing accounting
Bounded by entropy/information laws	Bounded by no-arbitrage under an ELMM

There is a precise martingale identity behind the analogy. If $\mathbb{Q} \approx \mathbb{P}$, define the density process

$$Z_t = \mathbb{E}_{\mathbb{P}}\left[\frac{d\mathbb{Q}}{d\mathbb{P}}\,\middle|\,\mathcal{F}_t\right]. \tag{6}$$

Then Z is a positive $\mathbb{P}$-martingale with $\mathbb{E}_{\mathbb{P}}[Z_T] = 1$. Writing $\Sigma_T = -\log Z_T$ gives the fluctuation-form identity

$$\mathbb{E}_{\mathbb{P}}\left[e^{-\Sigma_T}\right] = 1. \tag{7}$$

Equation (7) has the same normalization structure as an integral fluctuation relation such as Jarzynski's equality [9]. Here, however, Σ_T is a log-likelihood ratio between two financial measures, not physical entropy production.

An ELMM is also not the financial equivalent of thermodynamic detailed balance. The martingale property imposes zero conditional drift on discounted traded prices under $\mathbb{Q}$; detailed balance additionally imposes a time-reversal symmetry on a stationary transition law. The latter is strictly stronger.

B. Information Has a Financial Value, but Landauer Cost Does Not

Landauer's bound for erasing one bit at room temperature is approximately $k_B T \ln 2 \approx 2.9 \times 10^{-21}$ J per bit [10, 11]. Converting this into a "cost per trade" requires both the number of logically erased bits and an energy-price model. Even then, the bound is quantitatively negligible compared with spreads, impact, latency, and model risk; it is not the mechanism enforcing financial no-arbitrage.

The economically relevant information bound is instead expressed in growth units. In the canonical Kelly horse-race model with fair odds, the increase in optimal expected log wealth produced by side information Y about outcome X is the mutual information $I(X;Y)$ [12]. This result is model-specific, but it makes the demon analogy quantitative without confusing thermodynamic energy with money: exploitable information can increase optimal growth only because the joint law of signal and outcome is non-uniform.

V. THE COMBINATORIAL PERSPECTIVE

The optimization NFL theorem and the elementary prediction result needed for market-direction claims should be distinguished.

Theorem 4 (Optimization NFL, Wolpert–Macready)**.** *For finite search and cost spaces, when performance is averaged uniformly over all objective functions, any two non-revisiting black-box search algorithms have identical averaged performance.*

The theorem concerns oracle-based optimization over function spaces. It does not, by itself, imply a statement about self-financing gains on stochastic price paths. For binary directional prediction the relevant result is simpler and can be derived directly.

Proposition 5 (Uniform Binary Prediction)**.** *Let $X_1, \ldots, X_T$ be uniformly distributed on $\{0,1\}^T$. A causal deterministic predictor chooses $\widehat{X}_t = f_t(X_1, \ldots, X_{t-1})$. Its mean accuracy*

$$A_T = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\widehat{X}_t = X_t\}$$

satisfies $\mathbb{E}[A_T] = 1/2$. The same holds for a randomized predictor whose private randomization is independent of the next bit.

Proof. Conditional on every history $X_1, \ldots, X_{t-1}$, the next bit X_t is an independent fair bit. Therefore $\mathbb{P}(\widehat{X}_t = X_t \mid X_1, \ldots, X_{t-1}) = 1/2$. Linearity of expectation gives the result. Conditioning additionally on the predictor's private randomization proves the randomized case. □

Consequently, no predictor can achieve expected directional accuracy greater than $1/2$ under the uniform distribution over all binary sequences; in particular, none can exceed $1/2$ on every sequence. This is an NFL symmetry statement, not a model of actual markets. Financial sign sequences are not uniform in general, and a directional forecast does not determine trading profit because payoff sizes, financing, and stopping rules matter (Proposition 3).

The defensible practical conclusion is conditional: every successful predictor or strategy exploits non-uniform structure in the physical measure $\mathbb{P}$. Its edge is evidence about a restricted environment, not universal algorithmic superiority.

VI. THE COMPUTATIONAL PERSPECTIVE: ADAPTIVE COUNTERPATHS

We now discard the assumption of an exogenous price process and let the market react after observing the trader's current position. This is a worst-case, perfect-information model. Let a trading program M_σ be a *total* computable map from every finite computable price history H_t to a rational position w_t. Totality models the practical requirement that a deployed algorithm return an action before a deadline.

Theorem 6 (Computable Adaptive Counterpath)**.** *For every total deterministic computable strategy M_σ and every finite horizon T, there exists a positive computable price path P_{adv} on which its self-financing gain*

$$G_T = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t)$$

is non-positive. It is strictly negative if the strategy takes a nonzero position at least once.

Proof. Choose computable $P_0 > 0$ and rational $0 < \epsilon < 1$. At time t, compute $w_t = M_\sigma(H_t)$ and set

$$P_{t+1} = \begin{cases} P_t(1-\epsilon), & w_t > 0, \\ P_t(1+\epsilon), & w_t < 0, \\ P_t, & w_t = 0. \end{cases} \tag{8}$$

The resulting prices remain positive and computable. In each period, $w_t(P_{t+1} - P_t) \le 0$, with strict inequality when $w_t \ne 0$. Summing proves the claim. □

This is a simulate-and-oppose construction, not a reduction from the Halting Problem. Its quantifiers are $\forall M_\sigma\, \exists P_{\text{adv}}$: the counterpath may depend on the strategy. There need not be one finite path on which every strategy loses; for example, a permanently long and a permanently short position cannot both lose on the same one-step price move.

Corollary 7 (Maximin Value)**.** *In the finite-horizon game of Theorem 6, if the neutral strategy is allowed, then*

$$\sup_{M_\sigma} \inf_{P_{\text{adv}}} G_T(M_\sigma, P_{\text{adv}}) = 0.$$

Proof. The neutral strategy guarantees zero, while Theorem 6 gives every strategy a counterpath with gain at most zero. □

Proposition 8 (Private Randomization Does Not Create a Guarantee)**.** *Suppose a randomized trader's current coin toss is hidden from the environment. There still exists a passive computable distribution of next returns under which every integrable strategy has conditional expected gain zero.*

Proof. Let the next price increment be $+\epsilon P_t$ or $-\epsilon P_t$ with equal probability, independently of the trader's private draw. Conditional on the current information and position, $\mathbb{E}[w_t(P_{t+1} - P_t) \mid \mathcal{F}_t] = 0$. Summing gives expected gain zero. □

Randomization therefore prevents a code-observing opponent from mechanically opposing the unseen current draw, but it does not yield strictly positive expected gain in every environment.

A. The Algorithmic-Randomness Bridge

There is also a non-adaptive connection between computability and martingale finance. In the fair binary market, a nonnegative capital process $K\colon \{0,1\}^{<\mathbb{N}} \to \mathbb{R}_{\ge 0}$ is a martingale when

$$K(s) = \frac{K(s0) + K(s1)}{2}. \tag{9}$$

Theorem 9 (Ville's Inequality)**.** *If $K_0 = 1$ is a nonnegative martingale under the fair-coin measure, then for every $\lambda > 0$,*

$$\mathbb{P}\left(\sup_{n\ge 0} K_n \ge \lambda\right) \le \frac{1}{\lambda}. \tag{10}$$

Ville's inequality follows by stopping the nonnegative capital process when it first crosses λ and applying optional stopping, followed by a limit [13]. In particular, the set on which a fixed nonnegative martingale becomes unbounded has probability zero.

A binary sequence is *computably random* precisely when no computable nonnegative martingale succeeds on it by becoming unbounded [14, 15]. Since there are only countably many computable martingales, almost every fair-coin sequence is computably random. Thus a single typical passive path prevents unbounded enrichment by every computable fair-game betting system. This statement differs from Theorem 6: it is asymptotic, measure-one, and concerns unbounded success, whereas the adaptive counterpath gives a finite non-positive gain tailored to one strategy. Game-theoretic probability develops this pathwise betting viewpoint systematically [16].

B. What Rice's Theorem Does and Does Not Say

Rice's theorem states that every nontrivial extensional property of arbitrary partial computable functions is undecidable [17]. It therefore limits fully general program certification; even deciding whether arbitrary code is total is impossible in general [18]. It does *not* imply that the loss-producing inputs of a fixed total finite-horizon strategy are undecidable. On a specified finite computable path, such a program can be simulated and its gain calculated. No probability claim about market failure sets follows from Rice's theorem alone.

VII. ANALOGY TO SOCIAL CHOICE: SCOPE AND LIMITS

Arrow's Impossibility Theorem concerns social welfare functions on at least three alternatives. Under unrestricted preference profiles, collective rationality, Pareto unanimity, and independence of irrelevant alternatives, the aggregation rule must be dictatorial [19]. Its assumptions and objects differ from those of a trading model.

There is nevertheless a useful structural comparison:

Universal domain.: Arrow quantifies over all admissible preference profiles; pathwise trading universality quantifies over all admissible price paths.

Jointly incompatible demands.: Arrow's fairness and consistency requirements cannot all be met by a non-dictatorial rule. In trading, zero-cost strict profit on every path is incompatible with no-arbitrage.

No theorem transfer.: Neither result proves the other. The parallel is a shared warning that unrestricted domains make apparently natural universal requirements inconsistent.

Yanofsky shows how many self-referential arguments can be organized using fixed points and fixed-point-free transformations [20]. The response in Eq. (8) anti-aligns returns with every nonzero position, but it admits the neutral fixed point $w_t = 0$. That fixed point yields zero gain rather than strict profit. The relevant obstruction is therefore the absence of a *strictly profitable* fixed response, not the absence of every fixed point.

VIII. CASE STUDY: THE WHEEL OPTIONS STRATEGY

The Wheel sells a cash-secured put and, after assignment, sells a covered call against the acquired stock. Its economic exposure follows directly from put–call parity. Ignoring dividends and financing for the terminal-payoff identity,

$$S_T - (S_T - K)^+ = K - (K - S_T)^+. \tag{11}$$

The left-hand side is a covered-call payoff; the right-hand side is cash paying K plus a short put. After the corresponding time-zero costs are matched, a covered call and a cash-secured put are equivalent European payoffs. Rolling the Wheel therefore repeatedly maintains a put-like, short-downside-convexity exposure rather than creating directionless income.

A. Failure I: Catastrophic Drop

If the underlying crashes after put assignment, the short call liability decreases, which benefits the call writer, but the loss on the stock dominates and the combined covered-call position can suffer a large net loss. Equation (11) makes this explicit: below K, the payoff falls one-for-one with S_T. A reversed rally/crash pair can be used as the stress test in Criterion 1, but time reversal alone is not the proof; the payoff identity is.

B. Failure II: Persistent Decline

Repeated option premiums offset part of a persistent decline but do not remove the renewed short-put exposure. The loss is path-dependent rather than generally linear: strikes, maturities, volatility, assignment dates, and position sizing determine the sequence of gains.

C. Failure III: Violent Breakout and Benchmark Underperformance

In a large rally, shares may be called away at the call strike. The Wheel can still realize a positive absolute profit, so opportunity cost alone is not a counterexample to strict positive P&L. It is instead a failure relative to a buy-and-hold benchmark: Eq. (11) caps the terminal payoff at K. No-arbitrage constrains the price of this payoff under $\mathbb{Q}$; it does not say that its physical expected return must be zero.

D. Failure IV: Absorbing Ruin Under Capital Constraints

A fully cash-secured, unlevered cycle has bounded loss, but an underlying that falls to zero can consume almost all committed capital net of premium. Leverage makes the absorbing nature of ruin explicit. If wealth evolves as $W_{t+1} = W_t(1 + fR_{t+1})$ and an admissible return satisfies $1 + fR_{t+1} = 0$, then $W_{t+1} = 0$; a self-financing strategy cannot recover without an external capital injection. This is a wealth-dynamics fact, not a consequence of undecidability.

Table II is a payoff-based classification, not a claim that each trajectory has positive probability under every continuous price model. Corollary 2 excludes strict

TABLE II. Wheel failure modes and the mathematical mechanism illustrating each one.

Failure	**Trajectory Class**	**Relevant Result**	**Mechanism**
I: Crash	Large downside move	Eq. (11)	Put-like downside
II: Bleed	Persistent negative drift	Repeated payoff exposure	Premiums do not insure the stock
III: Breakout	Large upside move	Eq. (11)	Capped benchmark return
IV: Ruin	Capital exhaustion	Wealth recursion	Zero wealth is absorbing

positive discounted gain on all paths, but probabilities of particular failure classes require a specified physical measure.

IX. UNIVERSAL PORTFOLIOS ARE RELATIVE

The word "universal" also appears in a celebrated positive theorem. For price-relative vectors $x_t \in \mathbb{R}^m_{>0}$, a constant-rebalanced portfolio b has wealth

$$S_T(b) = \prod_{t=1}^{T} b^\top x_t.$$

Cover's universal portfolio is nonanticipating and, under the conditions of his theorem, has the same asymptotic exponential growth rate as the best constant-rebalanced portfolio chosen in hindsight on every individual bounded sequence [21]. In regret notation,

$$\frac{1}{T}\left[\log \max_b S_T(b) - \log S_T^{\mathrm{UP}}\right] \longrightarrow 0. \tag{12}$$

Equation (12) is a relative guarantee, not an absolute-profit guarantee. If every constant-rebalanced portfolio loses money on a path, the universal portfolio may lose as well while retaining vanishing per-period regret. Cover's result therefore does not contradict Corollary 2; it illustrates the strongest useful meaning of "universal" once the benchmark class is specified.

X. FAPP STRATEGIES AND THEIR HONEST SCOPE

While pathwise universal profit is precluded, strategies may be profitable *for all practical purposes* (FAPP, following Bell [22]) relative to a physical measure $\mathbb{P}$, a benchmark, and a finite horizon. Market making may earn spreads when adverse selection and inventory risk are controlled, and Kelly investing maximizes expected logarithmic growth when the relevant distribution is sufficiently well specified. Neither is an unconditional profit guarantee.

Economic equilibrium also leaves room for conditional edges. Grossman and Stiglitz show that perfectly informationally efficient markets are inconsistent with costly information acquisition: if prices revealed all information without reward, agents would not pay to produce that information [23]. Conversely, the Milgrom–Stokey no-trade theorem shows that, under common priors, common knowledge of rationality, risk aversion, and an initially Pareto-optimal allocation, private information alone cannot induce mutually agreeable non-null trade [24]. These are equilibrium boundaries, not substitutes for the FTAP.

Lo's Adaptive Markets Hypothesis [25] supplies a less idealized interpretation. As capital enters a successful strategy, competition, price impact, and strategic response may erode its edge. The market need not become the perfect adversary of Theorem 6; adaptation merely makes the physical distribution endogenous to widespread strategy use.

Automated execution can amplify feedback when many systems share triggers, funding constraints, or risk models. The 1987 portfolio-insurance cascade [26] and the 2010 Flash Crash [27] illustrate such interactions. Their relevance comes from market impact and coupled control rules, not from Rice's theorem.

XI. CONCLUSION

The impossibility of a universally winning zero-cost trading strategy is a direct consequence of no-arbitrage. Once gains are measured relative to a numeraire and admissibility and self-financing are stated explicitly, pathwise strict profit implies classical arbitrage. In a locally bounded semimartingale market satisfying NFLVR, an equivalent local martingale measure provides the standard expectation-based contradiction.

The complementary paradigms answer different questions. Uniform binary averaging fixes directional prediction accuracy at 1/2, while optional stopping shows that trade win rate can nevertheless be arbitrarily high without positive expected value. An adaptive computable environment can construct a finite counterpath against each total deterministic strategy. Under a passive fair measure, Ville's inequality and computable randomness show instead that typical paths prevent unbounded success by every computable martingale. None of these statements should be substituted for another.

The thermodynamic comparison is rigorous at the level of positive density martingales and fluctuation-form identities, not as an equivalence between martingale measures and detailed balance. The Wheel's put-like payoff, Cover's relative-regret guarantee, and information-economics results further show what remains possible: conditional growth, benchmark-relative performance, and transient informational rents. Any claim to work "in all market conditions" must therefore disclose its benchmark, financing, admissibility constraints, horizon, and assumptions about the market environment.

ACKNOWLEDGMENTS

Noson S. Yanofsky drew my attention to parallels between the present result, Arrow's Impossibility Theorem, and fixed-point formulations of self-reference.

This research was funded in whole or in part by the Austrian Science Fund (FWF) [Grant DOI: 10.55776/PIN5424624]. For open access purposes, the author has applied a CC BY public copyright license to any author accepted manuscript version arising from this submission. The author acknowledges TU Wien Bibliothek for financial support through its Open Access Funding Programme.

OpenAI Codex (GPT-5) was used to assist with literature organization, LaTeX editing, and checks of derivations. The author specified the scientific direction and prompts; model output retained in the manuscript was checked against the cited sources and explicit calculations. The author accepts full responsibility for the final text.

DATA AVAILABILITY

This is a purely mathematical work, and no data or software were created or analyzed in this study.

[1] L. Bachelier, Théorie de la spéculation, Annales Scientifiques de l'École Normale Supérieure **17**, 21 (1900).
[2] P. A. Samuelson, Proof that properly anticipated prices fluctuate randomly, Industrial Management Review **6**, 41 (1965).
[3] E. F. Fama, Efficient capital markets: A review of theory and empirical work, The Journal of Finance **25**, 383 (1970).
[4] D. H. Wolpert and W. G. Macready, No free lunch theorems for optimization, IEEE Transactions on Evolutionary Computation **1**, 67 (1997).
[5] C. Kardaras, Market viability via absence of arbitrage of the first kind, Finance and Stochastics **16**, 651 (2012).
[6] J. M. Harrison and S. R. Pliska, Martingales and stochastic integrals in the theory of continuous trading, Stochastic Processes and their Applications **11**, 215 (1981).
[7] F. Delbaen and W. Schachermayer, A general version of the fundamental theorem of asset pricing, Mathematische Annalen **300**, 463 (1994).
[8] J. L. Doob, *Stochastic Processes* (John Wiley & Sons, New York, 1953).
[9] C. Jarzynski, Nonequilibrium equality for free energy differences, Physical Review Letters **78**, 2690 (1997).
[10] R. Landauer, Irreversibility and heat generation in the computing process, IBM Journal of Research and Development **5**, 183 (1961).
[11] C. H. Bennett, The thermodynamics of computation—a review, International Journal of Theoretical Physics **21**, 905 (1982).
[12] J. L. Kelly, A new interpretation of information rate, The Bell System Technical Journal **35**, 917 (1956).
[13] J. Ville, *Étude critique de la notion de collectif*, Thèses de l'entre-deux-guerres No. 218 (Gauthier-Villars, Paris, 1939) pp. 1–124.
[14] C. P. Schnorr, *Zufälligkeit und Wahrscheinlichkeit*, Springer Lecture Notes in Mathematics, Vol. 218 (Springer, Berlin, 1971).
[15] R. G. Downey and D. R. Hirschfeldt, *Algorithmic Randomness and Complexity*, Theory and Applications of Computability (Springer, Berlin, 2010).
[16] G. Shafer and V. Vovk, *Probability and Finance: It's Only a Game!* (John Wiley & Sons, New York, 2001).
[17] H. G. Rice, Classes of recursively enumerable sets and their decision problems, Transactions of the American Mathematical Society **74**, 358 (1953).
[18] A. M. Turing, On computable numbers, with an application to the Entscheidungsproblem, Proceedings of the London Mathematical Society 2, **42**, 230 (1937).
[19] K. J. Arrow, *Social Choice and Individual Values*, 3rd ed. (Yale University Press, New Haven, CT, 2012) foreword to the Third Edition by Eric S. Maskin.
[20] N. S. Yanofsky, A universal approach to self-referential paradoxes, incompleteness and fixed points, Bulletin of Symbolic Logic **9**, 362 (2003), arXiv:math/0305282.
[21] T. M. Cover, Universal portfolios, Mathematical Finance **1**, 1 (1991).
[22] J. S. Bell, Against 'measurement', Physics World **3**, 33 (1990).
[23] S. J. Grossman and J. E. Stiglitz, On the impossibility of informationally efficient markets, The American Economic Review **70**, 393 (1980).
[24] P. Milgrom and N. Stokey, Information, trade and common knowledge, Journal of Economic Theory **26**, 17 (1982).
[25] A. W. Lo, The adaptive markets hypothesis: Market efficiency from an evolutionary perspective, The Journal of Portfolio Management **30**, 15 (2004).
[26] N. F. Brady, J. C. Cotting, R. G. Kirby, J. R. Opel, and H. M. Stein, *Report of the Presidential Task Force on Market Mechanisms* (U.S. Government Printing Office, Washington, D.C., 1988).
[27] A. Kirilenko, A. S. Kyle, M. Samadi, and T. Tuzun, The flash crash: High-frequency trading in an electronic market, The Journal of Finance **72**, 967 (2017).